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Parameters			Control Metrics			
α	k_p	k_i	Rise (f) ↑ / ↓	Settling (f) ↑ / ↓	OS (%) ↑ / ↓	SSE (ms) Step / Saw
0.83	2.0	0.33	17 / 10	15 / 8	14.2 / 39.4	1.90 / 1.71
0.85*	2.0	0.33	17 / 8	15 / 5	14.1 / 43.5	1.88 / 1.64
0.93	2.0	0.33	13 / 6	29 / 26	53.6 / 83.1	1.81 / 1.33
0.85	0.5	0.33	11 / 7	12 / 13	31.5 / 48.0	1.82 / 1.63
0.85	5.0	0.33	76 / 4	85 / 3	chat. / chat.	2.84 / 2.60
0.85	2.0	0.10	38 / 9	30 / 4	11.1 / 49.2	1.94 / 1.64
0.85	2.0	1.00	14 / 10	12 / 7	18.1 / 39.3	1.82 / 1.60

* Chosen default. Rise/Settling: causal MA-20, ±1.5 ms band. OS: causal MA-20. SSE: centered MA-30. † chat. = chattering regime (transition rate >60%); OS undefined.